

MAT 524: Functions of a Real Variable II

Homework II

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[[Problem No. 1]]:

Problem Statement

Prove that the function

$$f(x) = \begin{cases} x \sin(1/x) & x \in (0, 1] \\ 0 & x = 0 \end{cases}$$

does not have bounded variation on $[0, 1]$. Explain why this is a uniformly continuous function that is not absolutely continuous.

Solution

Let $n \in \mathbb{N}$ and let $P_n = \left\{0, \frac{2}{2n\pi}, \frac{2}{(2n-1)\pi}, \dots, \frac{2}{3\pi}, \frac{2}{2\pi}, 1\right\}$. Then, since $\sin\left(\frac{k\pi}{2}\right)$ is 0 for even k and 1 for odd k ,

$$V(f, P_n) = \frac{2}{(2n-1)\pi} + \frac{2}{(2n-1)\pi} + \frac{2}{(2n-3)\pi} + \frac{2}{(2n-3)\pi} + \dots + \sin(1) \quad (1)$$

$$= \frac{4}{(2n-1)\pi} + \frac{4}{(2n-3)\pi} + \dots + \sin(1). \quad (2)$$

Now, note that as n increases without bound $V(f, P_n)$ becomes $\sin(1) + \frac{4}{\pi} \sum_{m=1}^{\infty} \frac{1}{2m-1}$. However,

$$\sum_{m=1}^{\infty} \frac{1}{2m-1} > \sum_{m=1}^{\infty} \frac{1}{2m} \quad (3)$$

$$= \frac{1}{2} \sum_{m=1}^{\infty} \frac{1}{m} \quad (4)$$

$$= \infty. \quad (5)$$

Thus, $TV_{[0,1]}(f) = \sup \{V(f, P) \mid P \text{ a partition of } [0, 1]\} = \infty$, meaning f does not have bounded variation on $[0, 1]$.

To show that f is uniformly continuous on $[0, 1]$ (a compact interval), we simply need to show that it is continuous on $[0, 1]$ (because any function that is continuous on a compact interval is uniformly continuous on that interval by the Heine-Cantor Theorem). The only point at which the continuity of f is questionable is at $x = 0$. To show f is, indeed, continuous at $x = 0$, let $\varepsilon > 0$. Choose $\delta = \varepsilon$. Then whenever $|x - 0| = |x| < \delta$, we have

$$|f(x) - f(0)| = |x \sin(1/x) - 0| \tag{6}$$

$$= |x \sin(1/x)| \tag{7}$$

$$\leq |x| \tag{8}$$

$$< \delta \tag{9}$$

$$= \varepsilon. \tag{10}$$

Thus, f is continuous on $[0, 1]$ and, therefore, uniformly continuous on $[0, 1]$.

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[[Problem No. 2]]:

Problem Statement

Let $f : [a, b] \rightarrow \mathbb{R}$ be a function of bounded variation. For $x \in [a, b]$, define

$$V_f^+(x) = \sup_P \sum [f(x_{i+1}) - f(x_i)]^+,$$

where the sup is over all partitions P of $[a, x]$, and likewise

$$V_f^-(x) = \sup_P \sum [f(x_{i+1}) - f(x_i)]^-.$$

(a) Show that $V_f^+(x) - V_f^-(x) = f(x) - f(a)$ for all $x \in [a, b]$.

(b) Show that $V_f^+(x) + V_f^-(x) = TV_{[a,x]}(f)$ for all $x \in [a, b]$.

(c) Show that

$$\int_a^b |f'| \leq TV_{[a,b]}(f).$$

(d) Suppose now that f is absolutely continuous. Prove that the reverse inequality holds, and hence

$$\int_a^b |f'| = TV_{[a,b]}(f).$$

Solution

(a) Let $x_0 = a < x_1 < \dots < x_n = x$ be some partition of $[a, x]$. We have

$$\sum_{i=1}^n [f(x_{i+1}) - f(x_i)]^+ - \sum_{i=1}^n [f(x_{i+1}) - f(x_i)]^- = \sum_{i=1}^n [f(x_{i+1}) - f(x_i)] \quad (11)$$

$$= f(x_n) - f(x_0) \quad (12)$$

$$= f(x) - f(a), \quad (13)$$

due to the telescoping nature of the summation. Thus, we can write

$$\sum_{i=1}^{n-1} [f(x_{i+1}) - f(x_i)]^+ = \sum_{i=1}^{n-1} [f(x_{i+1}) - f(x_i)]^- + f(x) - f(a). \quad (14)$$

Since $f(x) - f(a)$ is a constant (if we fix x), we can take the supremum over all possible partitions of $[a, x]$ on both sides of Equation (14) to obtain

$$V_f^+(x) = V_f^-(x) + f(x) - f(a), \quad (15)$$

which, of course, implies that

$$V_f^+(x) - V_f^-(x) = f(x) - f(a), \quad (16)$$

as required.

(b) Again, let $x_0 = a < x_1 < \dots < x_n = x$ be some partition of $[a, x]$. We have

$$\sum_{i=1}^{n-1} [f(x_{i+1}) - f(x_i)]^+ + \sum_{i=1}^{n-1} [f(x_{i+1}) - f(x_i)]^- = \sum_{i=1}^{n-1} |f(x_{i+1}) - f(x_i)|. \quad (17)$$

We can rewrite Equation (17) as

$$2 \sum_{i=1}^{n-1} [f(x_{i+1}) - f(x_i)]^+ = \sum_{i=1}^{n-1} |f(x_{i+1}) - f(x_i)| + \sum_{i=1}^{n-1} [f(x_{i+1}) - f(x_i)]^+ - \sum_{i=1}^{n-1} [f(x_{i+1}) - f(x_i)]^- \quad (18)$$

$$= \sum_{i=1}^{n-1} |f(x_{i+1}) - f(x_i)| + f(x) - f(a). \quad (19)$$

We now take the supremum over all possible partitions of $[a, x]$ on both sides to obtain

$$2V_f^+(x) = TV_{[a,x]}(f) + f(x) - f(a) \quad (20)$$

$$= TV_{[a,x]}(f) + V_f^+(x) - V_f^-(x). \quad (21)$$

Rearranging Equation (21) yields

$$V_f^+(x) + V_f^-(x) = TV_{[a,x]}(f), \quad (22)$$

as required.

(c) By Part **(a)**, we can write $f(x) = V_f^+(x) - V_f^-(x) + f(a)$. Since $V_f^+(x)$ and $V_f^-(x)$ are monotone increasing functions on $[a, b]$, Lebesgue's Theorem guarantees that their derivatives exist almost everywhere

on $[a, b]$. Thus, the derivative of f exists almost everywhere on $[a, b]$, and

$$f' = V_f^{+'} - V_f^{-'}, \quad (23)$$

by the linearity of differentiation. Now, by Corollary 4, both $V_f^{+'}$ and $V_f^{-'}$ are integrable on $[a, b]$ and

$$\int_a^b V_f^{+'} \leq V_f^+(b) - V_f^+(a) \quad (24)$$

$$\int_a^b V_f^{-'} \leq V_f^-(b) - V_f^-(a). \quad (25)$$

Summing Equations (24) and (25), we obtain

$$\int_a^b (V_f^{+'} + V_f^{-'}) \leq V_f^+(b) + V_f^-(b) - (V_f^+(a) + V_f^-(a)) \quad (26)$$

$$= TV_{[a,b]}(f) - TV_{[a,a]}(f) \quad (27)$$

$$= TV_{[a,b]}(f), \quad (28)$$

by Part **(b)** and the linearity of the Lebesgue integral. Since V_f^+ and V_f^- are increasing functions, their derivatives are always positive. Thus,

$$TV_{[a,b]}(f) = \int_a^b (V_f^{+'} + V_f^{-'}) \quad (29)$$

$$= \int_a^b (|V_f^{+'}| + |V_f^{-'}|) \quad (30)$$

$$\geq \int_a^b |V_f^{+'} - V_f^{-'}| \quad (31)$$

$$= \int_a^b |f'|, \quad (32)$$

as needed.

(d) Let $x_0 = a < x_1 < \dots < x_n = b$ be some partition P of $[a, b]$. Let us now consider

$$V(f, P) = \sum_{i=1}^n |f(x_i) - f(x_{i-1})|, \quad (33)$$

the variation of f over this partition. Since f is absolutely continuous, we can rewrite the variation of f

over P using the Fundamental Theorem of Calculus; that is,

$$V(f, P) = \sum_{i=1}^n |f(x_i) - f(x_{i-1})| \quad (34)$$

$$= \sum_{i=1}^n \left| \int_{x_{i-1}}^{x_i} f' \right|. \quad (35)$$

Now,

$$\sum_{i=1}^n \left| \int_{x_{i-1}}^{x_i} f' \right| \leq \sum_{i=1}^n \int_{x_{i-1}}^{x_i} |f'| \quad (36)$$

$$= \int_a^b |f'|. \quad (37)$$

Thus, we have that

$$V(f, P) \leq \int_a^b |f'|. \quad (38)$$

Taking the supremum over all partitions of $[a, b]$ on both sides of (38) yields

$$TV_{[a,b]}(f) \leq \int_a^b |f'|, \quad (39)$$

as desired.

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[[Problem No. 3]]:

Problem Statement

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is absolutely continuous and monotone. Prove that for any set $D \subset [a, b]$ of measure 0, the image $f(D)$ also has measure 0

Solution

Let $D \subset [a, b]$ with $m(D) = 0$, and let $\varepsilon > 0$. Since f is absolutely continuous, there exists a $\delta > 0$ (depending on ε), which satisfies the requirements in the definition of absolute continuity. Furthermore, since $m(D) = 0$, we can construct an open cover $\mathcal{O} = \bigcup_{n=1}^{\infty} (a_n, b_n)$ of D satisfying $\sum_{n=1}^{\infty} [b_n - a_n] < \delta$. For this collection of intervals the absolute continuity of f guarantees that $\sum_{n=1}^{\infty} |f(b_n) - f(a_n)| < \varepsilon$. But, since f is monotone,

$$\sum_{n=1}^{\infty} |f(b_n) - f(a_n)| = m(f(\mathcal{O})). \quad (40)$$

Now, $f(D) \subseteq f(\mathcal{O})$, so $m(f(D)) \leq m(f(\mathcal{O}))$. Thus, $m(f(D)) < \varepsilon$. Since ε was arbitrary, this implies that $m(f(D)) = 0$.

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